

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1** The cell membrane controls which substances can enter and leave a cell.

The cell membrane is a phospholipid bilayer with proteins embedded in it.

- (a) (i) The shapes show the components of a phospholipid.

phosphate group



glycerol



fatty acid



covalent bond



Draw a diagram to show the structure of a phospholipid, using these shapes.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(ii) Which type of bond is found in this molecule?

(1)

- ☐ **A** ester
- ☐ **B** glycosidic
- ☐ **C** hydrogen
- ☐ **D** peptide

(b) (i) How does a nonpolar molecule move through a cell membrane, down a concentration gradient into a cell?

(1)

- ☐ **A** by active transport
- ☐ **B** by diffusion
- ☐ **C** by exocytosis
- ☐ **D** by osmosis

(ii) How many of the following statements are correct for the transport of a substance by endocytosis?

- transports substances out of the cell
- the cell membrane surrounds the substance
- can only transport substances down a concentration gradient

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

(Total for Question 1 = 5 marks)